Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details : TEAM C

Batch Number : 13
02/02/26

Name: Harsh Baldevprasad Mishra

Internship Duration: 8 Weeks

1. Project Title

NeuroFleetX — AI-Driven Urban Mobility Optimization System

2. Project Objective

To build a next-generation AI-driven platform that optimizes urban mobility and fleet operations for rental, transport, and smart city use cases. The system aims to:

- Enable real-time vehicle tracking and intelligent routing
- Implement predictive maintenance and dynamic fleet allocation
- Provide customer-centric service models
- Support EV fleets, ride-sharing models, and multi-modal transport networks
- Reduce traffic congestion — measured by the KPI: **average travel time per trip**

3. Project **description in detail**

NeuroFleetX is a next-generation AI-driven platform designed to optimize urban mobility and fleet operations for rental, transport, and smart city use cases. By leveraging artificial intelligence, IoT, and geospatial data, NeuroFleetX enables real-time vehicle tracking, intelligent routing, predictive maintenance, dynamic fleet allocation, and customer-centric service models.

The system integrates machine learning models to predict traffic conditions, optimize route plans, and monitor vehicle health. It also employs advanced dashboards and mobile-first interfaces for fleet managers and end-users. Designed for scalability, NeuroFleetX can be adapted for electric vehicle (EV) fleets, ride-sharing models, or multi-modal transport networks, offering both operational efficiency and sustainable mobility solutions for modern cities.

Module 1 — **Authentication & Role Management:**

Supports Admin, Manager, Driver, and Customer roles with secure login/registration using Spring Security and role-based dashboard redirects. Built using Spring Security and JWT authentication, each role gets a dedicated dashboard with relevant metric tiles (Bookings, Fleet, Users).

Module 2 — **Fleet Inventory & Vehicle Telemetry:**

Allows adding and updating vehicles, tracking real-time status (idle, in use, maintenance), and simulating telemetry data including speed, battery, fuel, and location. Each vehicle is displayed as a grid card with a location pin, status chip (Available / In Use / Needs Service), and a battery/fuel progress bar.

Module 3 — **AI Route & Load Optimization Engine:**

Provides dynamic route suggestions using AI — balancing shortest time, traffic, and energy. Integrates OpenStreetMap with a Dijkstra + ML-based ETA predictor. The map screen shows polyline routes with ETA cards and alternate colored route options.

Module 4 — **Predictive Maintenance & Health Analytics:**

Monitors engine, tire, fuel, battery, and mileage to predict the next maintenance date and alert the fleet manager. Vehicle health parameters are simulated and evaluated against thresholds. The dashboard includes a line graph (wear over time), an alert table (Vehicle ID, Issue, Action Needed), and a pie chart showing Healthy / Due / Critical status.

Module 5 — **Customer Booking & Smart Recommendations:**

Customers select vehicle type, location, and time. An AI recommendation engine suggests the best-fit vehicle based on past preferences. The booking screen includes filters (Type, Seats, EV/Non-EV),
Al-badged recommendation cards, and a booking calendar with per-slot pricing.

Module 6 — **Admin Dashboard & Urban Mobility Insights:**

Displays real-time fleet distribution, heatmaps of trip density, and downloadable reports (CSV, PDF). Built using Angular/React with Chart.js. Includes KPI cards (Total Fleet, Trips Today, Active Routes), a city heatmap, and a bar chart showing hourly rental activity.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project setup, requirement analysis, tech stack finalization	Set up development environment, created GitHub repository, finalized project scope and module breakdown, assigned roles to team members
Week 2	Module 1: Authentication & Role Management	Made the backend for registering the user and Log in Functionality. Implemented JWT token-based login system. Built the UI Auth pages for the same.
Week 3	Module 2: Fleet Inventory & Vehicle Telemetry	Modified backend logic. Built APIs and Vehicle CRUD system. Integrated simulated IoT telemetry for Location, speed, battery, and fuel.
Week 4	Module 2: RBAC and Vehicle Telemetry	Designed and routed pages for the users based on their roles. Created and stored table in the database called 'vehicles' and implemented functionality to add and manage vehicles (admin and manager only).
Week 5	Module 3: AI Route & Load Optimization Engine	Integrated OpenStreet Maps API, implemented single source shortest path Dijkstra's algorithm with ML-based ETA predictor, visualized polyline routes with alternate path options and traffic zones.
Week 6	Module 4: Predictive Maintenance & Health Analytics	Simulated vehicle health parameters (like low fuel), configured threshold-based alert triggers, built analytics dashboard with line graph, alert table, and maintenance pie chart.
Week 7	Module 5: Customer Booking & Smart Recommendations	Implemented shortest path + Eco path logic for both options. Implemented AI-based recommendation engine based on location, only show vehicles which are near to customer and which are 'IDLE'. Once booked, they become 'Running' and can't be booked until

		customer or driver marks the trip as ‘completed’.
Week 8	Module 6: data consistency throughout and integration of all modules	Made customer bookings manageable by driver, manager and admin. Ensured data remains consistent throughout the system, regardless of the role. Integrated final features and polished UI. Conducted end-to-end Integration testing.

5a. Key Milestones:

Milestone	Description	Date Achieved
Project Kickoff	Requirement analysis done, tech stack finalized	End of Week 1
Prototype/First Draft	Authentication and Fleet Inventory modules working	End of Week 3
Mid-Term Review	Route Optimization and Predictive Maintenance modules integrated	End of Week 5
Final Submission	All 6 modules complete, tested, report submitted	End of Week 8
Presentation	Live demo delivered to mentors and evaluators	End of Week 8

5b. Project execution details

The project was executed module by module across 8 weeks.

Module 1 involved designing login and registration logic, auth screens, implementing Spring Security with JWT-based token and authentication, and building role-specific dashboards for Admin, Manager, Driver, and Customer.

Module 2 involved creating a vehicle CRUD system, integrating dummy GPS coordinates and IoT sensor simulation, and adding status cards for each vehicle. Also, implementing role-specific features per user.

Module 3 involved integrating OpenStreetMap, viewing present vehicles on map UI, implementing a Dijkstra + ML-based ETA predictor, and visualizing routes with start, stop, ETA, and traffic zones on a map.

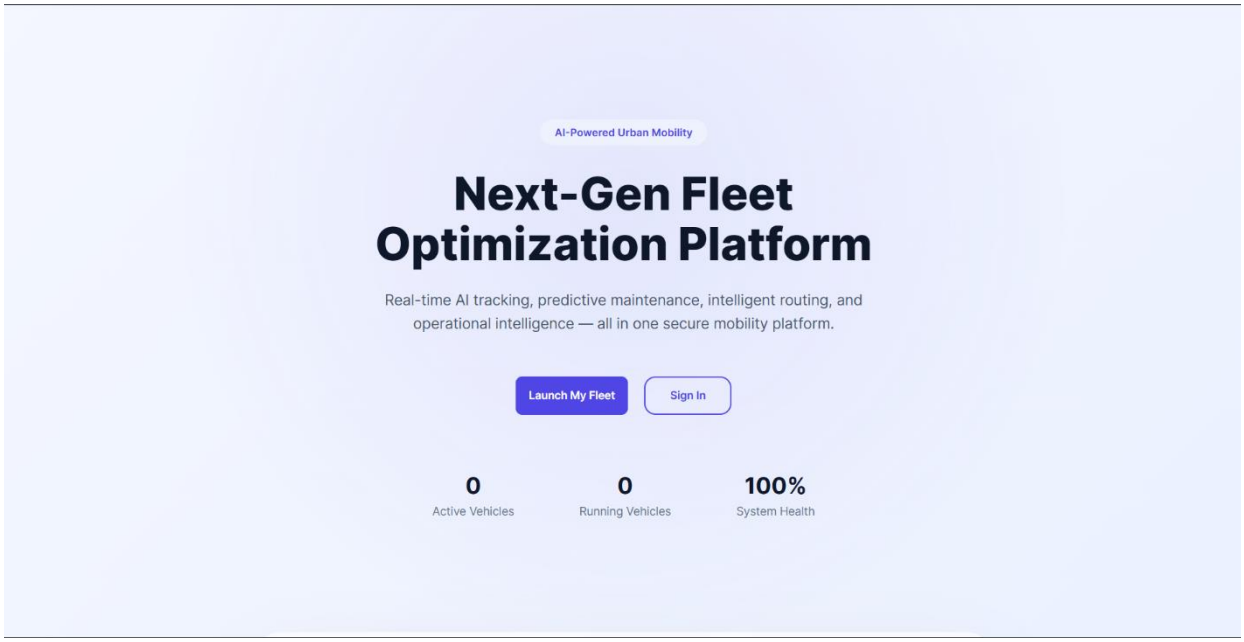
Module 4 involved simulating vehicle health parameters (engine, tire, fuel, battery, mileage), applying predefined thresholds to trigger warnings, and building an analytics dashboard along with Reports and Logs using line graphs, alert tables, and pie charts. Exported data to CSV using python pandas.

Module 5 involved building a customer UI to filter and search vehicles, implementing a recommendation engine using past data or dummy AI logic, and integrating a booking calendar with summary. Showing only available vehicles and assigning nearby vehicle to the customer. Also, reflect this change at driver side for his role. And let Admin and Manager control these two.

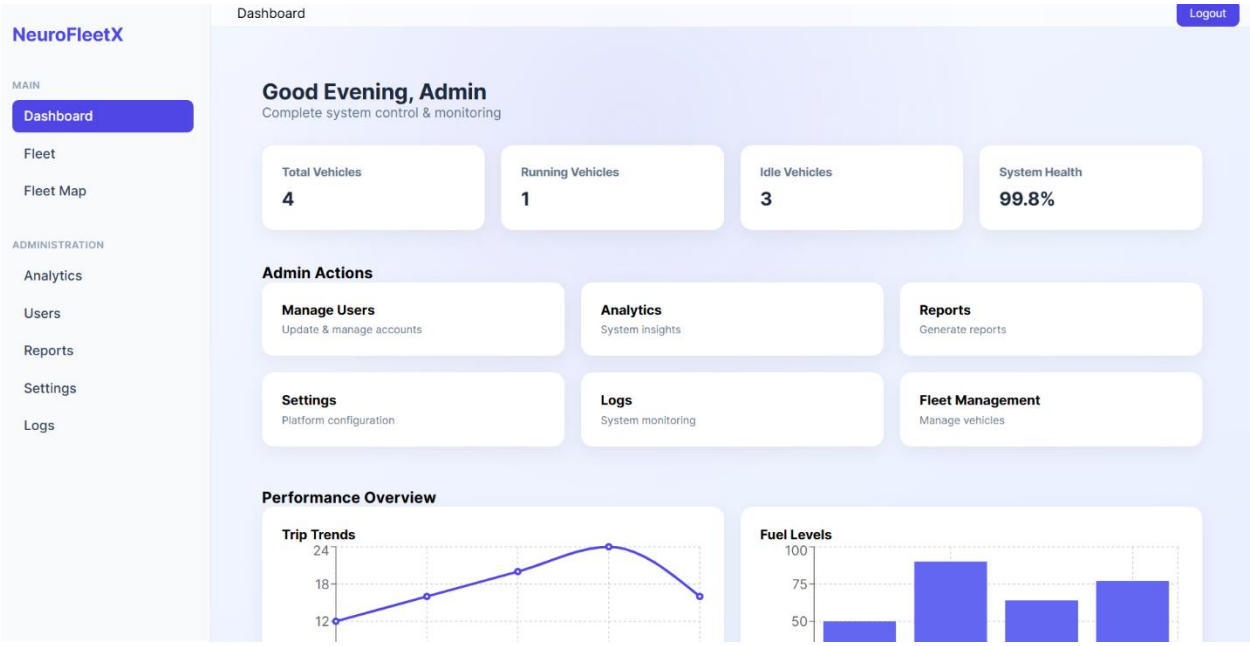
Module 6 involved creating an analytics dashboard using React and ReCharts, adding a map-based fleet distribution heatmap over the city layout, and including export/download buttons for CSV and PDF reports. Integrating the Modules, and adding validations to the system. Also, testing the system with load and boundary values.

6. Snapshots / Screenshots

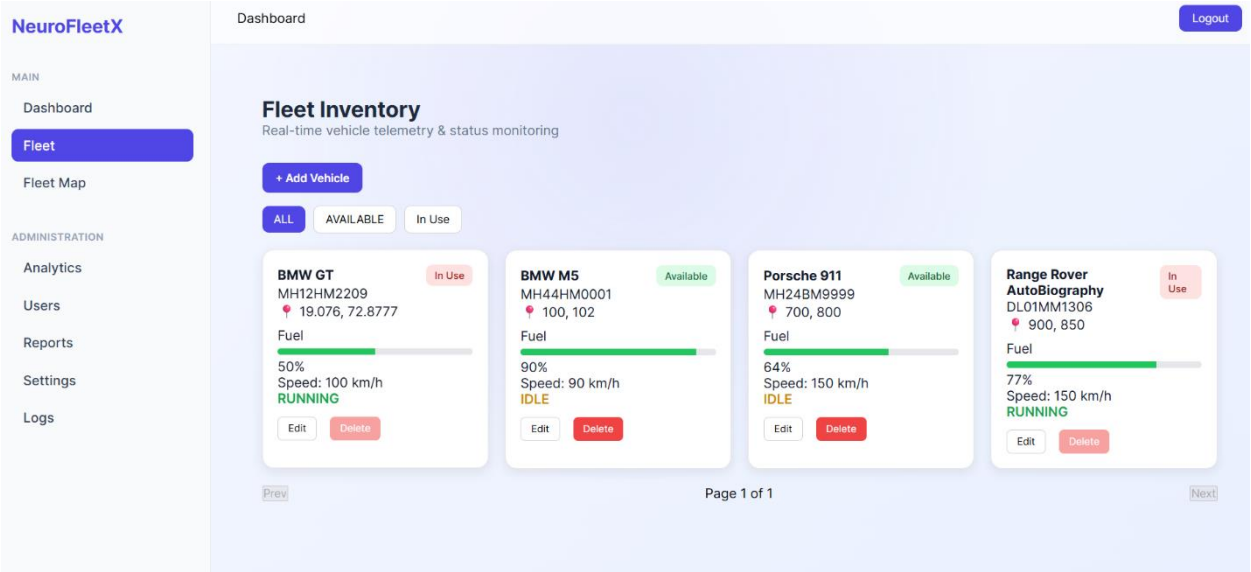
1. Landing Page:



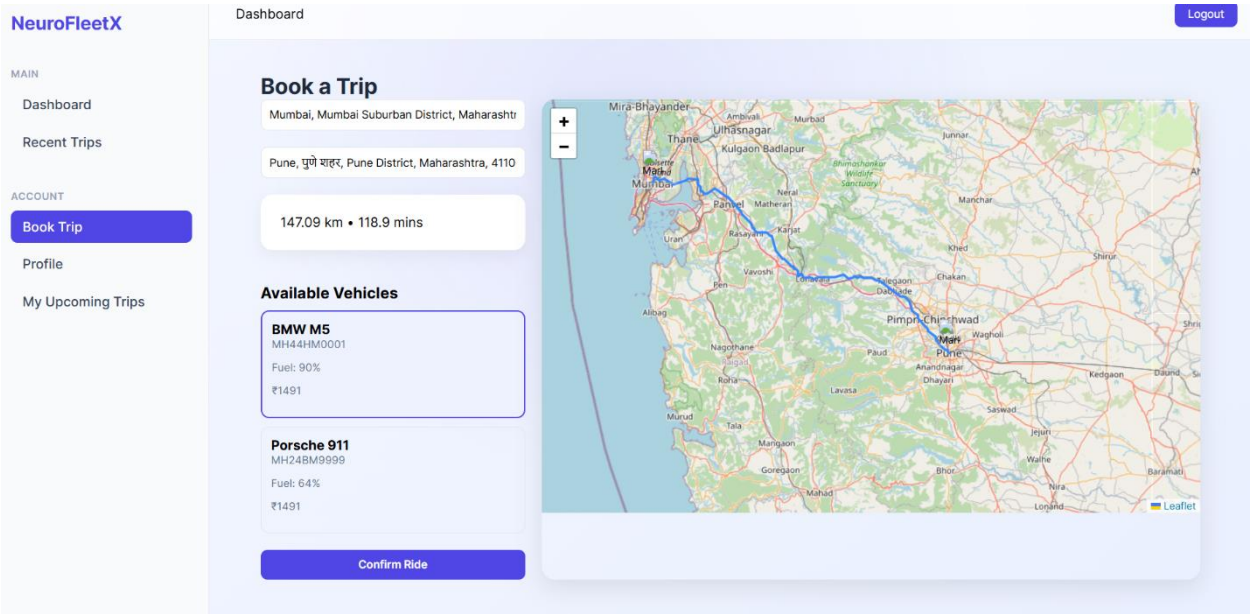
2. Admin Dashboard:



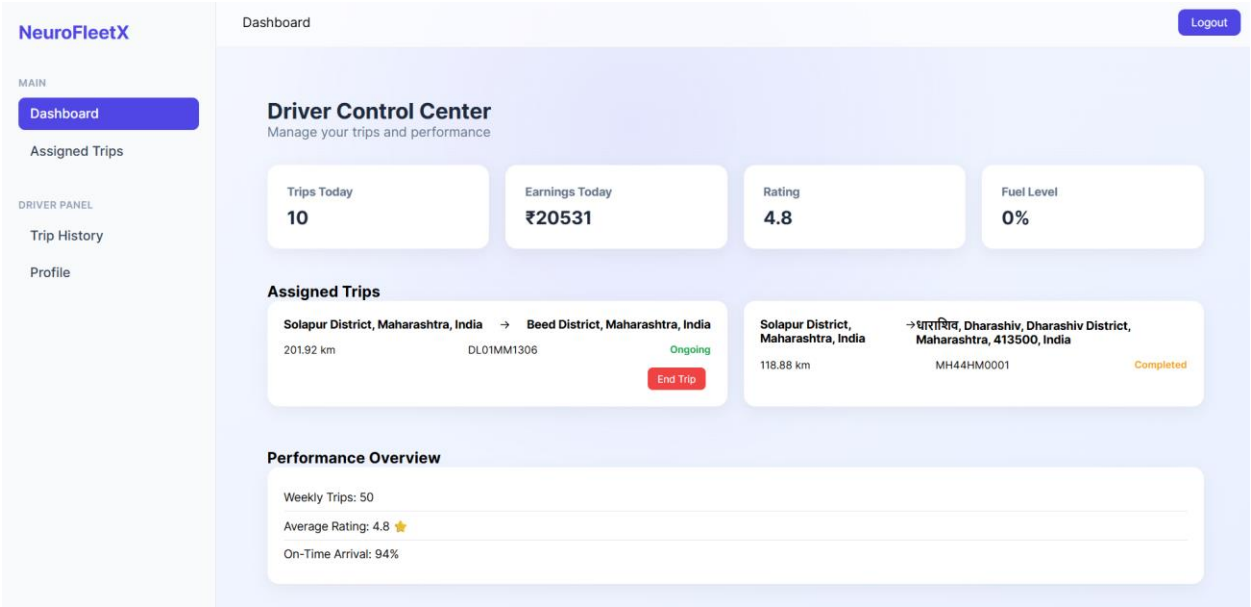
3. Fleet Inventory (Admin & Manager):



4. Customer Booking:



5. Driver Dashboard:



7. Challenges Faced

« Setting up JWT-based authentication with role-based access for 4 different user types
- Simulating realistic IoT telemetry data (GPS, battery, fuel) without physical hardware
- Integrating the Python Flask AI microservice with the Java Spring Boot backend

- Building an accurate ML-based ETA predictor with limited real traffic data
- Rendering real-time heatmaps on the admin dashboard with acceptable performance
- Defining and consistently tracking the KPI: average travel time per trip

8. Learnings & Skills Acquired

Technical:

- Full-stack development using Java Spring Boot and React 19
- JWT-based authentication and role management using Spring Security
- AI/ML integration: Dijkstra's algorithm and ML-based ETA prediction
- Geospatial mapping with OpenStreetMaps API and Leaflet.js
- Data visualization using Chart.js and Recharts
- Python Flask microservice development
- Real-time communication using WebSocket
- Database management with MySQL

Domain Knowledge:

- Fleet management systems and urban mobility challenges
- IoT telemetry simulation and predictive maintenance concepts
- Smart city architecture and multi-role enterprise application design
- KPI definition and measurement: average travel time per trip as a congestion metric

9. Testimonials from team

NeuroFleetX was unlike anything we had built in college. Integrating a Python AI microservice with a Spring Boot backend in the same project felt intimidating at first, but once we got it working, it was one of the most satisfying moments of the entire internship. I now feel genuinely confident working on full-stack enterprise systems.

10. Conclusion

NeuroFleetX was a comprehensive internship project that gave hands-on exposure to building a full-stack Al-powered platform for urban fleet management. The project successfully delivered all six modules — Authentication, Fleet Inventory, AI Route Optimization, Predictive Maintenance, Customer Booking, and Admin Analytics — using a modern tech stack of Spring Boot, Angular/React, Python Flask, MySQL, and Google Maps API.

The core goal of the project was to reduce traffic congestion, with average travel time per trip as the primary KPI. Every design and development decision across the 8 weeks was guided by this objective. The internship aligned closely with our academic background while introducing us to real-world challenges in AI integration, system design, and agile development practices.

11. Acknowledgements

We extend our heartfelt gratitude to our internship mentor for their constant guidance, technical support, and patience throughout the eight weeks of this program. Their clarity in explaining concepts like KPI definition, microservice architecture, and AI integration helped us bridge the gap between academic knowledge and real-world development.

We sincerely thank the organization for giving us the opportunity to work on a meaningful and challenging project like NeuroFleetX. The resources, tools, and collaborative environment provided to us made it possible to build a platform we are genuinely proud of.

We also thank our fellow interns and team members for their dedication, teamwork, and the many late debugging sessions that turned into some of our best learning moments. This internship has been a defining experience in our journey as software developers.